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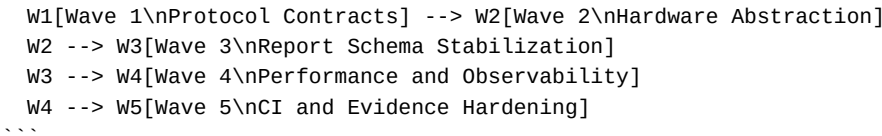
Executive Summary

Metric	Value
---	---
Rust files scanned	60
Functions detected	848
Critical risk functions	66
High risk functions	52
Medium risk functions	310
Low risk functions	420

Visual Roadmap

```mermaid

flowchart LR



## 1. Matriz de Risco por Funcao Critica

<details>

<summary>Expand full critical and high risk matrix</summary>

| Risco    | Funcao                                    | Local                      | Linha | Justificativa                                                        | Testes Minimos                                                                                                                                                                                                                          |
|----------|-------------------------------------------|----------------------------|-------|----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| critical | handle_uds_serial_request                 | src/bin/cat_comm_bridge.rs | 101   | impacts diagnostic protocol contract and ECU session state           | cargo check ; cargo check --features vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test vendor_bridge_e2e ; powershell -NoProfile -ExecutionPolicy Bypass -File scripts/run_windows_template_e2e.ps1 |
| critical | missing_for_live_flash                    | src/io/hw.rs               | 49    | impacts firmware write path, transfer integrity, and product outcome | cargo check ; cargo check --features vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test vendor_bridge_e2e ; powershell -NoProfile -ExecutionPolicy Bypass -File scripts/run_windows_template_e2e.ps1 |
| critical | declared_capabilities                     | src/io/hw.rs               | 231   | impacts hardware transport abstraction and live connectivity         | cargo check --features vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test vendor_bridge_e2e ; powershell -NoProfile -ExecutionPolicy Bypass -File scripts/run_windows_template_e2e.ps1               |
| critical | open_real_adapter                         | src/io/hw.rs               | 298   | impacts hardware transport abstraction and live connectivity         | cargo check --features vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test vendor_bridge_e2e ; powershell -NoProfile -ExecutionPolicy Bypass -File scripts/run_windows_template_e2e.ps1               |
| critical | probe_live_adapter                        | src/io/hw.rs               | 324   | impacts hardware transport abstraction and live connectivity         | cargo check --features vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test vendor_bridge_e2e ; powershell -NoProfile -ExecutionPolicy Bypass -File scripts/run_windows_template_e2e.ps1               |
| critical | append_flash_audit                        | src/io/live_runner.rs      | 484   | impacts firmware write path, transfer integrity, and product outcome | cargo check ; cargo check --features vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test vendor_bridge_e2e ; powershell -NoProfile -ExecutionPolicy Bypass -File scripts/run_windows_template_e2e.ps1 |
| critical | run_flash_preflight                       | src/io/live_runner.rs      | 749   | impacts firmware write path, transfer integrity, and product outcome | cargo check ; cargo check --features vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test vendor_bridge_e2e ; powershell -NoProfile -ExecutionPolicy Bypass -File scripts/run_windows_template_e2e.ps1 |
| critical | j1939_uds_req_id                          | src/io/live_runner.rs      | 816   | impacts diagnostic protocol contract and ECU session state           | cargo check --features vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test vendor_bridge_e2e ; powershell -NoProfile -ExecutionPolicy Bypass -File scripts/run_windows_template_e2e.ps1               |
| critical | is_uds_response_id                        | src/io/live_runner.rs      | 820   | impacts diagnostic protocol contract and ECU session state           | cargo check --features vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test vendor_bridge_e2e ; powershell -NoProfile -ExecutionPolicy Bypass -File scripts/run_windows_template_e2e.ps1               |
| critical | recv_can_uds_response                     | src/io/live_runner.rs      | 1046  | impacts diagnostic protocol contract and ECU session state           | cargo check --features vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test vendor_bridge_e2e ; powershell -NoProfile -ExecutionPolicy Bypass -File scripts/run_windows_template_e2e.ps1               |
| critical | send_uds_and_wait                         | src/io/live_runner.rs      | 1164  | impacts diagnostic protocol contract and ECU session state           | cargo check --features vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test vendor_bridge_e2e ; powershell -NoProfile -ExecutionPolicy Bypass -File scripts/run_windows_template_e2e.ps1               |
| critical | live_flash_ecm_firmware                   | src/io/live_runner.rs      | 1286  | impacts firmware write path, transfer integrity, and product outcome | cargo check ; cargo check --features vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test vendor_bridge_e2e ; powershell -NoProfile -ExecutionPolicy Bypass -File scripts/run_windows_template_e2e.ps1 |
| critical | uds_response_id_matches_expected_target   | src/io/live_runner.rs      | 1631  | impacts diagnostic protocol contract and ECU session state           | cargo check ; cargo check --features vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test vendor_bridge_e2e ; powershell -NoProfile -ExecutionPolicy Bypass -File scripts/run_windows_template_e2e.ps1 |
| critical | uds_response_id_rejects_wrong_pf_ps_or_sa | src/io/live_runner.rs      | 1639  | impacts diagnostic protocol contract and ECU session state           | cargo check ; cargo check --features vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test vendor_bridge_e2e ; powershell -NoProfile -ExecutionPolicy Bypass -File scripts/run_windows_template_e2e.ps1 |

ssion state | cargo check ; cargo check --features vendor-windows ; cargo test --workspace ; cargo test --features ven  
t vendor\_bridge\_e2e ; powershell -NoProfile -ExecutionPolicy Bypass -File scripts/run\_windows\_template\_e2e.ps1 |  
| critical | run\_all\_phases | src/io/production\_program.rs | 77 | impacts release gate and conformance reporting | car  
check --features vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test vendor\_bridge\_e  
NoProfile -ExecutionPolicy Bypass -File scripts/run\_windows\_template\_e2e.ps1 |  
| critical | run\_sim\_identity\_check | src/io/production\_program.rs | 601 | impacts release gate and conformance report  
; cargo check --features vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test vendor  
ershell -NoProfile -ExecutionPolicy Bypass -File scripts/run\_windows\_template\_e2e.ps1 |  
| critical | build\_conformance\_summary | src/io/production\_program.rs | 613 | impacts release gate and conformance rep  
eck ; cargo check --features vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test ven  
powershell -NoProfile -ExecutionPolicy Bypass -File scripts/run\_windows\_template\_e2e.ps1 |  
| critical | run\_uds\_runtime\_self\_test | src/io/production\_program.rs | 744 | impacts diagnostic protocol contract and  
e | cargo check ; cargo check --features vendor-windows ; cargo test --workspace ; cargo test --features vendor-window  
ridge\_e2e ; powershell -NoProfile -ExecutionPolicy Bypass -File scripts/run\_windows\_template\_e2e.ps1 |  
| critical | supports | src/io/vendor\_cat\_comm.rs | 30 | impacts hardware transport abstraction and live connectivity  
argo check --features vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test vendor\_bri  
ell -NoProfile -ExecutionPolicy Bypass -File scripts/run\_windows\_template\_e2e.ps1 |  
| critical | open | src/io/vendor\_cat\_comm.rs | 88 | impacts hardware transport abstraction and live connectivity | ca  
check --features vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test vendor\_bridge\_  
-NoProfile -ExecutionPolicy Bypass -File scripts/run\_windows\_template\_e2e.ps1 |  
| critical | ensure\_connected | src/io/vendor\_cat\_comm.rs | 154 | impacts hardware transport abstraction and live conn  
check ; cargo check --features vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test v  
; powershell -NoProfile -ExecutionPolicy Bypass -File scripts/run\_windows\_template\_e2e.ps1 |  
| critical | send\_bridge\_request | src/io/vendor\_cat\_comm.rs | 164 | impacts hardware transport abstraction and live c  
go check ; cargo check --features vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --tes  
2e ; powershell -NoProfile -ExecutionPolicy Bypass -File scripts/run\_windows\_template\_e2e.ps1 |  
| critical | read\_bridge\_response | src/io/vendor\_cat\_comm.rs | 182 | impacts hardware transport abstraction and live  
rgo check ; cargo check --features vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --te  
e2e ; powershell -NoProfile -ExecutionPolicy Bypass -File scripts/run\_windows\_template\_e2e.ps1 |  
| critical | append\_diag\_record | src/io/vendor\_cat\_comm.rs | 201 | impacts hardware transport abstraction and live co  
o check ; cargo check --features vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test  
e ; powershell -NoProfile -ExecutionPolicy Bypass -File scripts/run\_windows\_template\_e2e.ps1 |  
| critical | init | src/io/vendor\_cat\_comm.rs | 233 | impacts hardware transport abstraction and live connectivity | c  
o check --features vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test vendor\_bridge  
-NoProfile -ExecutionPolicy Bypass -File scripts/run\_windows\_template\_e2e.ps1 |  
| critical | read\_frame | src/io/vendor\_cat\_comm.rs | 246 | impacts hardware transport abstraction and live connectivi  
; cargo check --features vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test vendor\_  
rshell -NoProfile -ExecutionPolicy Bypass -File scripts/run\_windows\_template\_e2e.ps1 |  
| critical | try\_read\_frame | src/io/vendor\_cat\_comm.rs | 261 | impacts hardware transport abstraction and live connec  
eck ; cargo check --features vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test ven  
powershell -NoProfile -ExecutionPolicy Bypass -File scripts/run\_windows\_template\_e2e.ps1 |  
| critical | send\_frame | src/io/vendor\_cat\_comm.rs | 273 | impacts hardware transport abstraction and live connectivi  
; cargo check --features vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test vendor\_  
rshell -NoProfile -ExecutionPolicy Bypass -File scripts/run\_windows\_template\_e2e.ps1 |  
| critical | close | src/io/vendor\_cat\_comm.rs | 286 | impacts hardware transport abstraction and live connectivity |  
go check --features vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test vendor\_bridg  
l -NoProfile -ExecutionPolicy Bypass -File scripts/run\_windows\_template\_e2e.ps1 |  
| critical | adapter\_info | src/io/vendor\_cat\_comm.rs | 305 | impacts hardware transport abstraction and live connecti  
k ; cargo check --features vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test vendo  
wershell -NoProfile -ExecutionPolicy Bypass -File scripts/run\_windows\_template\_e2e.ps1 |  
| critical | resolve\_bridge\_exe | src/io/vendor\_cat\_comm.rs | 318 | impacts hardware transport abstraction and live co  
o check ; cargo check --features vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test  
e ; powershell -NoProfile -ExecutionPolicy Bypass -File scripts/run\_windows\_template\_e2e.ps1 |  
| critical | map\_bridge\_error | src/io/vendor\_cat\_comm.rs | 347 | impacts hardware transport abstraction and live conn  
check ; cargo check --features vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test v  
; powershell -NoProfile -ExecutionPolicy Bypass -File scripts/run\_windows\_template\_e2e.ps1 |  
| critical | bridge\_info\_from\_response | src/io/vendor\_cat\_comm.rs | 368 | impacts hardware transport abstraction and  
| cargo check ; cargo check --features vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows  
idge\_e2e ; powershell -NoProfile -ExecutionPolicy Bypass -File scripts/run\_windows\_template\_e2e.ps1 |  
| critical | ensure\_bridge\_compatibility | src/io/vendor\_cat\_comm.rs | 375 | impacts hardware transport abstraction an  
ty | cargo check ; cargo check --features vendor-windows ; cargo test --workspace ; cargo test --features vendor-windo  
bridge\_e2e ; powershell -NoProfile -ExecutionPolicy Bypass -File scripts/run\_windows\_template\_e2e.ps1 |  
| critical | now\_ms | src/io/vendor\_cat\_comm.rs | 413 | impacts hardware transport abstraction and live connectivity |  
rgo check --features vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test vendor\_brid  
ll -NoProfile -ExecutionPolicy Bypass -File scripts/run\_windows\_template\_e2e.ps1 |  
| critical | engineering\_oil\_catalog | src/leak\_physics.rs | 496 | impacts runtime behavior and downstream callers | c  
o check --features vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test vendor\_bridge  
-NoProfile -ExecutionPolicy Bypass -File scripts/run\_windows\_template\_e2e.ps1 |  
| critical | uds\_process\_sa | src/main.rs | 6236 | impacts diagnostic protocol contract and ECU session state | cargo  
ck --features vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test vendor\_bridge\_e2e

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rofile -ExecutionPolicy Bypass -File scripts/run_windows_template_e2e.ps1 |
| critical | uds_send_and_log | src/main.rs | 6246 | impacts diagnostic protocol contract and ECU session state | cargo
heck --features vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test vendor_bridge_e2
oProfile -ExecutionPolicy Bypass -File scripts/run_windows_template_e2e.ps1 |
| critical | run_uds_flash_pipeline | src/main.rs | 6278 | impacts firmware write path, transfer integrity, and produc
| cargo check ; cargo check --features vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows
idge_e2e ; powershell -NoProfile -ExecutionPolicy Bypass -File scripts/run_windows_template_e2e.ps1 |
| critical | tab_uds | src/main.rs | 9040 | impacts diagnostic protocol contract and ECU session state | cargo check ;
atures vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test vendor_bridge_e2e ; power
-ExecutionPolicy Bypass -File scripts/run_windows_template_e2e.ps1 |
| critical | fmt | src/uds.rs | 42 | impacts runtime behavior and downstream callers | cargo check ; cargo check --fea
ows ; cargo test --workspace ; cargo test --features vendor-windows --test vendor_bridge_e2e ; powershell -NoProfile
ypass -File scripts/run_windows_template_e2e.ps1 |
| critical | new | src/uds.rs | 237 | impacts runtime behavior and downstream callers | cargo check ; cargo check --fe
dows ; cargo test --workspace ; cargo test --features vendor-windows --test vendor_bridge_e2e ; powershell -NoProfile
Bypass -File scripts/run_windows_template_e2e.ps1 |
| critical | tick | src/uds.rs | 276 | impacts runtime behavior and downstream callers | cargo check ; cargo check --f
ndows ; cargo test --workspace ; cargo test --features vendor-windows --test vendor_bridge_e2e ; powershell -NoProfile
Bypass -File scripts/run_windows_template_e2e.ps1 |
| critical | process | src/uds.rs | 293 | impacts runtime behavior and downstream callers | cargo check ; cargo check
-windows ; cargo test --workspace ; cargo test --features vendor-windows --test vendor_bridge_e2e ; powershell -NoProf
icy Bypass -File scripts/run_windows_template_e2e.ps1 |
| critical | svc_session_control | src/uds.rs | 320 | impacts runtime behavior and downstream callers | cargo check ;
tures vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test vendor_bridge_e2e ; powers
ExecutionPolicy Bypass -File scripts/run_windows_template_e2e.ps1 |
| critical | svc_ecu_reset | src/uds.rs | 353 | impacts runtime behavior and downstream callers | cargo check ; cargo
vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test vendor_bridge_e2e ; powershell -
ionPolicy Bypass -File scripts/run_windows_template_e2e.ps1 |
| critical | svc_clear_dtc | src/uds.rs | 395 | impacts runtime behavior and downstream callers | cargo check ; cargo
vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test vendor_bridge_e2e ; powershell -
ionPolicy Bypass -File scripts/run_windows_template_e2e.ps1 |
| critical | svc_read_dtc | src/uds.rs | 428 | impacts runtime behavior and downstream callers | cargo check ; cargo c
endor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test vendor_bridge_e2e ; powershell -N
onPolicy Bypass -File scripts/run_windows_template_e2e.ps1 |
| critical | svc_read_data_by_id | src/uds.rs | 504 | impacts runtime behavior and downstream callers | cargo check ;
tures vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test vendor_bridge_e2e ; powers
ExecutionPolicy Bypass -File scripts/run_windows_template_e2e.ps1 |
| critical | svc_security_access | src/uds.rs | 554 | impacts diagnostic protocol contract and ECU session state | car
check --features vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test vendor_bridge_e
NoProfile -ExecutionPolicy Bypass -File scripts/run_windows_template_e2e.ps1 |
| critical | svc_write_data_by_id | src/uds.rs | 640 | impacts runtime behavior and downstream callers | cargo check ;
atures vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test vendor_bridge_e2e ; power
-ExecutionPolicy Bypass -File scripts/run_windows_template_e2e.ps1 |
| critical | svc_routine_control | src/uds.rs | 684 | impacts runtime behavior and downstream callers | cargo check ;
tures vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test vendor_bridge_e2e ; powers
ExecutionPolicy Bypass -File scripts/run_windows_template_e2e.ps1 |
| critical | svc_request_download | src/uds.rs | 733 | impacts firmware write path, transfer integrity, and production
argo check ; cargo check --features vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --t
_e2e ; powershell -NoProfile -ExecutionPolicy Bypass -File scripts/run_windows_template_e2e.ps1 |
| critical | svc_transfer_data | src/uds.rs | 767 | impacts firmware write path, transfer integrity, and production ga
o check ; cargo check --features vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test
e ; powershell -NoProfile -ExecutionPolicy Bypass -File scripts/run_windows_template_e2e.ps1 |
| critical | svc_request_transfer_exit | src/uds.rs | 809 | impacts firmware write path, transfer integrity, and produ
e | cargo check ; cargo check --features vendor-windows ; cargo test --workspace ; cargo test --features vendor-window
ridge_e2e ; powershell -NoProfile -ExecutionPolicy Bypass -File scripts/run_windows_template_e2e.ps1 |
| critical | svc_tester_present | src/uds.rs | 830 | impacts runtime behavior and downstream callers | cargo check ; c
ures vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test vendor_bridge_e2e ; powersh
xecutionPolicy Bypass -File scripts/run_windows_template_e2e.ps1 |
| critical | record_dtc_set | src/uds.rs | 850 | impacts runtime behavior and downstream callers | cargo check ; cargo
vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test vendor_bridge_e2e ; powershell
tionPolicy Bypass -File scripts/run_windows_template_e2e.ps1 |
| critical | nrc | src/uds.rs | 879 | impacts runtime behavior and downstream callers | cargo check ; cargo check --fe
dows ; cargo test --workspace ; cargo test --features vendor-windows --test vendor_bridge_e2e ; powershell -NoProfile
Bypass -File scripts/run_windows_template_e2e.ps1 |
| critical | log | src/uds.rs | 883 | impacts runtime behavior and downstream callers | cargo check ; cargo check --fe
dows ; cargo test --workspace ; cargo test --features vendor-windows --test vendor_bridge_e2e ; powershell -NoProfile
Bypass -File scripts/run_windows_template_e2e.ps1 |
| critical | service_name | src/uds.rs | 905 | impacts runtime behavior and downstream callers | cargo check ; cargo c
endor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test vendor_bridge_e2e ; powershell -N
onPolicy Bypass -File scripts/run_windows_template_e2e.ps1 |
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| critical | dtc\_to\_3bytes | src/uds.rs | 926 | impacts runtime behavior and downstream callers | cargo check ; cargo test --workspace ; cargo test --features vendor-windows --test vendor\_bridge\_e2e ; powershell -File scripts/run\_windows\_template\_e2e.ps1 |

| critical | session\_default\_relocks\_security | src/uds.rs | 939 | impacts diagnostic protocol contract and ECU session integrity, and production gate outcome | cargo check ; cargo check --features vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test vendor\_bridge\_e2e ; powershell -NoProfile -ExecutionPolicy Bypass -File scripts/run\_windows\_template\_e2e.ps1 |

| critical | transfer\_exit\_rejects\_incomplete\_download | src/uds.rs | 949 | impacts firmware write path, transfer integrity, and production gate outcome | cargo check ; cargo check --features vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test vendor\_bridge\_e2e ; powershell -NoProfile -ExecutionPolicy Bypass -File scripts/run\_windows\_template\_e2e.ps1 |

| critical | bin\_path | tests/vendor\_bridge\_e2e.rs | 9 | impacts hardware transport abstraction and live connectivity | cargo check --features vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test vendor\_bridge\_e2e ; powershell -NoProfile -ExecutionPolicy Bypass -File scripts/run\_windows\_template\_e2e.ps1 |

| critical | unique\_temp\_dir | tests/vendor\_bridge\_e2e.rs | 22 | impacts hardware transport abstraction and live connectivity | cargo check --features vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test vendor\_bridge\_e2e ; powershell -NoProfile -ExecutionPolicy Bypass -File scripts/run\_windows\_template\_e2e.ps1 |

| critical | cat\_comm\_bridge\_probe\_and\_live\_phases\_flash\_e2e | tests/vendor\_bridge\_e2e.rs | 33 | impacts firmware write path, transfer integrity, and production gate outcome | cargo check ; cargo check --features vendor-windows ; cargo test --workspace ; cargo test --features vendor-windows --test vendor\_bridge\_e2e ; powershell -NoProfile -ExecutionPolicy Bypass -File scripts/run\_windows\_template\_e2e.ps1 |

| high | from\_path | src/io/allowlist.rs | 33 | impacts runtime behavior and downstream callers | cargo check ; cargo test --test io\_mock |

| high | is\_allowed | src/io/allowlist.rs | 42 | impacts runtime behavior and downstream callers | cargo check ; cargo test --test io\_mock |

| high | per\_rule\_rate\_limit | src/io/allowlist.rs | 46 | impacts runtime behavior and downstream callers | cargo check ; cargo test --workspace ; cargo test --test io\_mock |

| high | validate | src/io/allowlist.rs | 53 | impacts runtime behavior and downstream callers | cargo check ; cargo test --test io\_mock |

| high | rule\_matches\_frame | src/io/allowlist.rs | 95 | impacts runtime behavior and downstream callers | cargo check ; cargo test --workspace ; cargo test --test io\_mock |

| high | parse\_u32\_hex\_or\_dec | src/io/allowlist.rs | 137 | impacts runtime behavior and downstream callers | cargo check ; cargo test --workspace ; cargo test --test io\_mock |

| high | parse\_mask | src/io/allowlist.rs | 146 | impacts runtime behavior and downstream callers | cargo check ; cargo test --test io\_mock |

| high | match\_hex\_pattern | src/io/allowlist.rs | 150 | impacts runtime behavior and downstream callers | cargo check ; cargo test --workspace ; cargo test --test io\_mock |

| high | can\_rule\_matches\_mask | src/io/allowlist.rs | 182 | impacts runtime behavior and downstream callers | cargo check ; cargo test --workspace ; cargo test --test io\_mock |

| high | serial\_rule\_matches\_pattern | src/io/allowlist.rs | 209 | impacts runtime behavior and downstream callers | cargo check ; cargo test --workspace ; cargo test --test io\_mock |

| high | write\_intent\_enabled | src/io/hw.rs | 223 | impacts hardware transport abstraction and live connectivity | cargo check ; cargo test --workspace ; cargo test --test io\_mock |

| high | connect\_ecm | src/io/live\_runner.rs | 174 | impacts runtime behavior and downstream callers | cargo check ; cargo test --workspace ; cargo test --test io\_mock |

| high | live\_preflight\_ecm | src/io/live\_runner.rs | 793 | impacts runtime behavior and downstream callers | cargo check ; cargo test --workspace ; cargo test --test io\_mock |

| high | check\_live\_channel\_policy | src/io/live\_runner.rs | 1216 | impacts runtime behavior and downstream callers | cargo check ; cargo test --workspace ; cargo test --test io\_mock |

| high | on\_rate\_limited | src/io/metrics.rs | 37 | impacts runtime behavior and downstream callers | cargo check ; cargo test --test io\_mock |

| high | inject\_disconnect | src/io/mock.rs | 32 | impacts runtime behavior and downstream callers | cargo check ; cargo test --test io\_mock |

| high | clear\_disconnect | src/io/mock.rs | 36 | impacts runtime behavior and downstream callers | cargo check ; cargo test --test io\_mock |

| high | read\_frame | src/io/mock.rs | 61 | impacts runtime behavior and downstream callers | cargo check ; cargo test --test io\_mock |

| high | try\_read\_frame | src/io/mock.rs | 89 | impacts runtime behavior and downstream callers | cargo check ; cargo test --test io\_mock |

| high | send\_frame | src/io/mock.rs | 114 | impacts runtime behavior and downstream callers | cargo check ; cargo test --test io\_mock |

| high | default | src/io/production\_program.rs | 69 | impacts release gate and conformance reporting | cargo check ; cargo test --workspace ; cargo test --test io\_mock |

| high | run\_sim\_baseline\_check | src/io/production\_program.rs | 568 | impacts release gate and conformance reporting | cargo check ; cargo test --workspace ; cargo test --test io\_mock |

| high | run\_sim\_connectivity\_check | src/io/production\_program.rs | 587 | impacts release gate and conformance reporting | cargo check ; cargo test --workspace ; cargo test --test io\_mock |

| high | run\_sim\_preflight\_check | src/io/production\_program.rs | 731 | impacts release gate and conformance reporting | cargo check ; cargo test --workspace ; cargo test --test io\_mock |

| high | run\_sim\_stress\_check | src/io/production\_program.rs | 763 | impacts release gate and conformance reporting | cargo check ; cargo test --workspace ; cargo test --test io\_mock |

| high | hex\_sha | src/io/production\_program.rs | 784 | impacts release gate and conformance reporting | cargo check ;

kspace ; cargo test --test io\_mock |  
| high | now\_ms | src/io/production\_program.rs | 792 | impacts release gate and conformance reporting | cargo check ;  
space ; cargo test --test io\_mock |  
| high | new | src/io/rate\_limiter.rs | 13 | impacts runtime behavior and downstream callers | cargo check ; cargo tes  
argo test --test io\_mock |  
| high | try\_take | src/io/rate\_limiter.rs | 23 | impacts runtime behavior and downstream callers | cargo check ; carg  
e ; cargo test --test io\_mock |  
| high | refill | src/io/rate\_limiter.rs | 33 | impacts runtime behavior and downstream callers | cargo check ; cargo  
; cargo test --test io\_mock |  
| high | new | src/io/rate\_limiter.rs | 49 | impacts runtime behavior and downstream callers | cargo check ; cargo tes  
argo test --test io\_mock |  
| high | check\_can | src/io/rate\_limiter.rs | 57 | impacts runtime behavior and downstream callers | cargo check ; ca  
ce ; cargo test --test io\_mock |  
| high | check\_serial | src/io/rate\_limiter.rs | 70 | impacts runtime behavior and downstream callers | cargo check ;  
space ; cargo test --test io\_mock |  
| high | limiter\_blocks\_when\_empty | src/io/rate\_limiter.rs | 80 | impacts runtime behavior and downstream callers | c  
o test --workspace ; cargo test --test io\_mock |  
| high | aggregate\_group\_report | src/leak\_physics.rs | 73 | impacts evidence schema consumed by automation and approv  
; cargo test --workspace ; cargo test --test io\_mock |  
| high | export\_calibration\_report\_json | src/leak\_physics.rs | 1654 | impacts evidence schema consumed by automation  
argo check ; cargo test --workspace ; cargo test --test io\_mock |  
| high | export\_calibration\_report\_csv | src/leak\_physics.rs | 1670 | impacts evidence schema consumed by automation a  
rgo check ; cargo test --workspace ; cargo test --test io\_mock |  
| high | report\_captures\_pressure\_band\_and\_safe\_window | src/leak\_physics.rs | 2030 | impacts evidence schema consumed  
d approvals | cargo check ; cargo test --workspace ; cargo test --test io\_mock |  
| high | calibration\_mode\_ingests\_csv\_and\_exports\_reports | src/leak\_physics.rs | 2107 | impacts evidence schema consu  
and approvals | cargo check ; cargo test --workspace ; cargo test --test io\_mock |  
| high | export\_leak\_report\_json | src/lib.rs | 1053 | impacts evidence schema consumed by automation and approvals |  
go test --workspace ; cargo test --test io\_mock |  
| high | export\_leak\_report\_csv | src/lib.rs | 1060 | impacts evidence schema consumed by automation and approvals | c  
o test --workspace ; cargo test --test io\_mock |  
| high | export\_leak\_calibration\_report\_json | src/lib.rs | 1118 | impacts evidence schema consumed by automation and  
check ; cargo test --workspace ; cargo test --test io\_mock |  
| high | export\_leak\_calibration\_report\_csv | src/lib.rs | 1126 | impacts evidence schema consumed by automation and a  
check ; cargo test --workspace ; cargo test --test io\_mock |  
| high | pick\_save\_path | src/main.rs | 542 | impacts runtime behavior and downstream callers | cargo check ; cargo te  
cargo test --test io\_mock |  
| high | leak\_ascii\_cad | src/main.rs | 6360 | impacts runtime behavior and downstream callers | cargo check ; cargo t  
cargo test --test io\_mock |  
| high | fmt | src/v2x\_telematics.rs | 322 | impacts runtime behavior and downstream callers | cargo check ; cargo tes  
argo test --test io\_mock |  
| high | write\_test\_allowlist | tests/io\_mock.rs | 10 | impacts write authorization and operational safety | cargo che  
-workspace ; cargo test --test io\_mock |  
| high | mock\_adapter\_allows\_write\_when\_policy\_satisfied | tests/io\_mock.rs | 97 | impacts hardware transport abstract  
ectivity | cargo check ; cargo test --workspace ; cargo test --test io\_mock |  
| high | mock\_adapter\_dry\_run\_logs\_without\_physical\_tx | tests/io\_mock.rs | 125 | impacts hardware transport abstracti  
ctivity | cargo check ; cargo test --workspace ; cargo test --test io\_mock |  
| high | mock\_adapter\_rate\_limit\_blocks\_burst | tests/io\_mock.rs | 157 | impacts hardware transport abstraction and li  
cargo check ; cargo test --workspace ; cargo test --test io\_mock |  
| high | allowlist\_parse\_mask\_accepts\_hex | tests/io\_mock.rs | 189 | impacts write authorization and operational safet  
cargo test --workspace ; cargo test --test io\_mock |  
| high | e2e\_hydraulic\_plus\_ac\_stress\_generates\_reports | tests/leak\_system\_integration.rs | 4 | impacts evidence sche  
tomation and approvals | cargo check ; cargo test --workspace ; cargo test --test io\_mock |

</details>

## 2. Se Mudar X Entao Quebra Y

<details>

<summary>Expand full X->Y impact table with line references</summary>

| X alterado | Onde | Risco | Y afetado | Linhas de referencia | Efeito esperado |  
|---|---|---|---|---|---|  
| handle\_uds\_serial\_request | src/bin/cat\_comm\_bridge.rs | critica | chamadas diretas nao detectadas por varredura est  
at\_comm\_bridge.rs:101 | impacto local com potencial efeito indireto via fluxo de controle |  
| missing\_for\_live\_flash | src/io/hw.rs | critica | src/io/production\_program.rs ; src/io/production\_program.rs ; src/  
gram.rs | src/io/hw.rs:49 ; src/io/production\_program.rs:94 ; src/io/production\_program.rs:705 ; src/io/production\_pro  
e quebrar protocolo, gate de producao ou fluxo de flash |  
| declared\_capabilities | src/io/hw.rs | critica | src/io/production\_program.rs | src/io/hw.rs:231 ; src/io/production  
pode quebrar protocolo, gate de producao ou fluxo de flash |

| open\_real\_adapter | src/io/hw.rs | critica | src/io/live\_runner.rs ; src/io/live\_runner.rs ; src/io/live\_runner.rs ;  
er.rs ; src/io/live\_runner.rs ; src/io/live\_runner.rs | src/io/hw.rs:298 ; src/io/live\_runner.rs:141 ; src/io/live\_run  
io/live\_runner.rs:277 ; src/io/live\_runner.rs:786 ; src/io/live\_runner.rs:804 ; src/io/live\_runner.rs:1252 | pode queb  
te de producao ou fluxo de flash |  
| probe\_live\_adapter | src/io/hw.rs | critica | src/bin/simulator\_cli.rs ; tests/vendor\_bridge\_e2e.rs | src/io/hw.rs:3  
lator\_cli.rs:36 ; tests/vendor\_bridge\_e2e.rs:60 | pode quebrar protocolo, gate de producao ou fluxo de flash |  
| append\_flash\_audit | src/io/live\_runner.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/  
:484 | impacto local com potencial efeito indireto via fluxo de controle |  
| run\_flash\_preflight | src/io/live\_runner.rs | critica | chamadas diretas nao detectadas por varredura estatica | src  
s:749 | impacto local com potencial efeito indireto via fluxo de controle |  
| j1939\_uds\_req\_id | src/io/live\_runner.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/io  
16 | impacto local com potencial efeito indireto via fluxo de controle |  
| is\_uds\_response\_id | src/io/live\_runner.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/  
:820 | impacto local com potencial efeito indireto via fluxo de controle |  
| recv\_can\_uds\_response | src/io/live\_runner.rs | critica | chamadas diretas nao detectadas por varredura estatica | s  
.rs:1046 | impacto local com potencial efeito indireto via fluxo de controle |  
| send\_uds\_and\_wait | src/io/live\_runner.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/i  
1164 | impacto local com potencial efeito indireto via fluxo de controle |  
| live\_flash\_ecm\_firmware | src/io/live\_runner.rs | critica | src/io/production\_program.rs ; src/main.rs | src/io/live  
src/io/production\_program.rs:291 ; src/main.rs:9291 | pode quebrar protocolo, gate de producao ou fluxo de flash |  
| uds\_response\_id\_matches\_expected\_target | src/io/live\_runner.rs | critica | chamadas diretas nao detectadas por varr  
src/io/live\_runner.rs:1631 | impacto local com potencial efeito indireto via fluxo de controle |  
| uds\_response\_id\_rejects\_wrong\_pf\_ps\_or\_sa | src/io/live\_runner.rs | critica | chamadas diretas nao detectadas por va  
| src/io/live\_runner.rs:1639 | impacto local com potencial efeito indireto via fluxo de controle |  
| run\_all\_phases | src/io/production\_program.rs | critica | src/bin/simulator\_cli.rs ; tests/vendor\_bridge\_e2e.rs | sr  
rogram.rs:77 ; src/bin/simulator\_cli.rs:73 ; tests/vendor\_bridge\_e2e.rs:64 | pode quebrar protocolo, gate de producao  
|  
| run\_sim\_identity\_check | src/io/production\_program.rs | critica | chamadas diretas nao detectadas por varredura esta  
duction\_program.rs:601 | impacto local com potencial efeito indireto via fluxo de controle |  
| build\_conformance\_summary | src/io/production\_program.rs | critica | chamadas diretas nao detectadas por varredura e  
production\_program.rs:613 | impacto local com potencial efeito indireto via fluxo de controle |  
| run\_uds\_runtime\_self\_test | src/io/production\_program.rs | critica | chamadas diretas nao detectadas por varredura e  
production\_program.rs:744 | impacto local com potencial efeito indireto via fluxo de controle |  
| supports | src/io/vendor\_cat\_comm.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/io/ven  
0 | impacto local com potencial efeito indireto via fluxo de controle |  
| open | src/io/vendor\_cat\_comm.rs | critica | src/io/hw.rs ; src/io/hw.rs ; src/io/hw.rs ; src/io/hw.rs ; src/io/repl  
play.rs | src/io/vendor\_cat\_comm.rs:88 ; src/io/hw.rs:303 ; src/io/hw.rs:310 ; src/io/hw.rs:315 ; src/io/hw.rs:391 ; s  
1 ; src/io/replay.rs:42 | pode quebrar protocolo, gate de producao ou fluxo de flash |  
| ensure\_connected | src/io/vendor\_cat\_comm.rs | critica | chamadas diretas nao detectadas por varredura estatica | sr  
omm.rs:154 | impacto local com potencial efeito indireto via fluxo de controle |  
| send\_bridge\_request | src/io/vendor\_cat\_comm.rs | critica | chamadas diretas nao detectadas por varredura estatica |  
t\_comm.rs:164 | impacto local com potencial efeito indireto via fluxo de controle |  
| read\_bridge\_response | src/io/vendor\_cat\_comm.rs | critica | chamadas diretas nao detectadas por varredura estatica  
at\_comm.rs:182 | impacto local com potencial efeito indireto via fluxo de controle |  
| append\_diag\_record | src/io/vendor\_cat\_comm.rs | critica | chamadas diretas nao detectadas por varredura estatica |  
\_comm.rs:201 | impacto local com potencial efeito indireto via fluxo de controle |  
| init | src/io/vendor\_cat\_comm.rs | critica | src/io/hw.rs ; src/io/hw.rs ; src/io/live\_runner.rs ; src/io/live\_runne  
e\_runner.rs ; src/io/live\_runner.rs | src/io/vendor\_cat\_comm.rs:233 ; src/io/hw.rs:287 ; src/io/hw.rs:326 ; src/io/liv  
src/io/live\_runner.rs:183 ; src/io/live\_runner.rs:288 ; src/io/live\_runner.rs:787 | pode quebrar protocolo, gate de p  
de flash |  
| read\_frame | src/io/vendor\_cat\_comm.rs | critica | src/io/hw.rs ; src/io/live\_runner.rs ; src/io/live\_runner.rs ; sr  
rs ; src/io/live\_runner.rs ; src/io/live\_runner.rs | src/io/vendor\_cat\_comm.rs:246 ; src/io/hw.rs:288 ; src/io/live\_ru  
/io/live\_runner.rs:213 ; src/io/live\_runner.rs:302 ; src/io/live\_runner.rs:702 ; src/io/live\_runner.rs:897 | pode queb  
te de producao ou fluxo de flash |  
| try\_read\_frame | src/io/vendor\_cat\_comm.rs | critica | src/io/hw.rs ; src/io/mock.rs ; src/io/serial\_adapter.rs ; sr  
apter.rs | src/io/vendor\_cat\_comm.rs:261 ; src/io/hw.rs:289 ; src/io/mock.rs:89 ; src/io/serial\_adapter.rs:61 ; src/io  
r.rs:80 | pode quebrar protocolo, gate de producao ou fluxo de flash |  
| send\_frame | src/io/vendor\_cat\_comm.rs | critica | src/io/hw.rs ; src/io/live\_runner.rs ; src/io/live\_runner.rs ; sr  
c/io/serial\_adapter.rs ; src/io/socketcan\_adapter.rs | src/io/vendor\_cat\_comm.rs:273 ; src/io/hw.rs:290 ; src/io/live\_  
rc/io/live\_runner.rs:840 ; src/io/mock.rs:114 ; src/io/serial\_adapter.rs:69 ; src/io/socketcan\_adapter.rs:89 | pode qu  
gate de producao ou fluxo de flash |  
| close | src/io/vendor\_cat\_comm.rs | critica | src/io/hw.rs ; src/io/hw.rs ; src/io/live\_runner.rs ; src/io/live\_runn  
ve\_runner.rs ; src/io/live\_runner.rs | src/io/vendor\_cat\_comm.rs:286 ; src/io/hw.rs:291 ; src/io/hw.rs:330 ; src/io/li  
; src/io/live\_runner.rs:168 ; src/io/live\_runner.rs:203 ; src/io/live\_runner.rs:227 | pode quebrar protocolo, gate de  
de flash |  
| adapter\_info | src/io/vendor\_cat\_comm.rs | critica | src/io/hw.rs ; src/io/hw.rs | src/io/vendor\_cat\_comm.rs:305 ; s  
src/io/hw.rs:328 | pode quebrar protocolo, gate de producao ou fluxo de flash |  
| resolve\_bridge\_exe | src/io/vendor\_cat\_comm.rs | critica | chamadas diretas nao detectadas por varredura estatica |  
\_comm.rs:318 | impacto local com potencial efeito indireto via fluxo de controle |  
| map\_bridge\_error | src/io/vendor\_cat\_comm.rs | critica | chamadas diretas nao detectadas por varredura estatica | sr

omm.rs:347 | impacto local com potencial efeito indireto via fluxo de controle |  
| bridge\_info\_from\_response | src/io/vendor\_cat\_comm.rs | critica | chamadas diretas nao detectadas por varredura esta  
dor\_cat\_comm.rs:368 | impacto local com potencial efeito indireto via fluxo de controle |  
| ensure\_bridge\_compatibility | src/io/vendor\_cat\_comm.rs | critica | chamadas diretas nao detectadas por varredura es  
endor\_cat\_comm.rs:375 | impacto local com potencial efeito indireto via fluxo de controle |  
| now\_ms | src/io/vendor\_cat\_comm.rs | critica | src/bin/cat\_comm\_bridge.rs ; src/bin/cat\_comm\_bridge.rs ; src/io/hw.r  
; src/io/live\_runner.rs ; src/io/live\_runner.rs | src/io/vendor\_cat\_comm.rs:413 ; src/bin/cat\_comm\_bridge.rs:97 ; src/  
ge.rs:362 ; src/io/hw.rs:353 ; src/io/hw.rs:400 ; src/io/live\_runner.rs:200 ; src/io/live\_runner.rs:373 | pode quebrar  
de producao ou fluxo de flash |  
| engineering\_oil\_catalog | src/leak\_physics.rs | critica | chamadas diretas nao detectadas por varredura estatica | s  
s:496 | impacto local com potencial efeito indireto via fluxo de controle |  
| uds\_process\_sa | src/main.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/main.rs:6236  
m potencial efeito indireto via fluxo de controle |  
| uds\_send\_and\_log | src/main.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/main.rs:6246  
com potencial efeito indireto via fluxo de controle |  
| run\_uds\_flash\_pipeline | src/main.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/main.r  
local com potencial efeito indireto via fluxo de controle |  
| tab\_uds | src/main.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/main.rs:9040 | impact  
cial efeito indireto via fluxo de controle |  
| fmt | src/uds.rs | critica | src/adas.rs ; src/autonomous.rs ; src/autonomous.rs ; src/autonomous.rs ; src/autonomou  
mous.rs | src/uds.rs:42 ; src/adas.rs:56 ; src/autonomous.rs:16 ; src/autonomous.rs:39 ; src/autonomous.rs:75 ; src/au  
src/autonomous.rs:173 | pode quebrar protocolo, gate de producao ou fluxo de flash |  
| new | src/uds.rs | critica | src/adas.rs ; src/autonomous.rs ; src/autonomous.rs ; src/autonomous.rs ; src/autonomou  
mous.rs | src/uds.rs:237 ; src/adas.rs:67 ; src/autonomous.rs:185 ; src/autonomous.rs:190 ; src/autonomous.rs:230 ; sr  
45 ; src/autonomous.rs:570 | pode quebrar protocolo, gate de producao ou fluxo de flash |  
| tick | src/uds.rs | critica | src/autonomous.rs ; src/autonomous.rs ; src/autonomous.rs ; src/autonomous.rs ; src/au  
/bin/simulator\_cli.rs | src/uds.rs:276 ; src/autonomous.rs:245 ; src/autonomous.rs:546 ; src/autonomous.rs:584 ; src/a  
; src/autonomous.rs:630 ; src/bin/simulator\_cli.rs:125 | pode quebrar protocolo, gate de producao ou fluxo de flash |  
| process | src/uds.rs | critica | src/can\_gateway.rs ; src/io/production\_program.rs ; src/io/production\_program.rs ;  
\_program.rs ; src/io/production\_program.rs ; src/io/production\_program.rs | src/uds.rs:293 ; src/can\_gateway.rs:210 ;  
\_program.rs:603 ; src/io/production\_program.rs:604 ; src/io/production\_program.rs:746 ; src/io/production\_program.rs:7  
ction\_program.rs:748 | pode quebrar protocolo, gate de producao ou fluxo de flash |  
| svc\_session\_control | src/uds.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/uds.rs:320  
com potencial efeito indireto via fluxo de controle |  
| svc\_ecu\_reset | src/uds.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/uds.rs:353 | imp  
tencial efeito indireto via fluxo de controle |  
| svc\_clear\_dtc | src/uds.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/uds.rs:395 | imp  
tencial efeito indireto via fluxo de controle |  
| svc\_read\_dtc | src/uds.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/uds.rs:428 | impa  
encial efeito indireto via fluxo de controle |  
| svc\_read\_data\_by\_id | src/uds.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/uds.rs:504  
com potencial efeito indireto via fluxo de controle |  
| svc\_security\_access | src/uds.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/uds.rs:554  
com potencial efeito indireto via fluxo de controle |  
| svc\_write\_data\_by\_id | src/uds.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/uds.rs:64  
com potencial efeito indireto via fluxo de controle |  
| svc\_routine\_control | src/uds.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/uds.rs:684  
com potencial efeito indireto via fluxo de controle |  
| svc\_request\_download | src/uds.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/uds.rs:73  
com potencial efeito indireto via fluxo de controle |  
| svc\_transfer\_data | src/uds.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/uds.rs:767 |  
m potencial efeito indireto via fluxo de controle |  
| svc\_request\_transfer\_exit | src/uds.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/uds.  
local com potencial efeito indireto via fluxo de controle |  
| svc\_tester\_present | src/uds.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/uds.rs:830  
om potencial efeito indireto via fluxo de controle |  
| record\_dtc\_set | src/uds.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/uds.rs:850 | im  
otencial efeito indireto via fluxo de controle |  
| nrc | src/uds.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/uds.rs:879 | impacto local  
eito indireto via fluxo de controle |  
| log | src/uds.rs | critica | src/can\_gateway.rs ; src/can\_gateway.rs ; src/can\_gateway.rs ; src/nvm.rs | src/uds.rs:  
eway.rs:110 ; src/can\_gateway.rs:128 ; src/can\_gateway.rs:281 ; src/nvm.rs:32 | pode quebrar protocolo, gate de produc  
ash |  
| service\_name | src/uds.rs | critica | src/main.rs | src/uds.rs:905 ; src/main.rs:6252 | pode quebrar protocolo, gate  
luxo de flash |  
| dtc\_to\_3bytes | src/uds.rs | critica | chamadas diretas nao detectadas por varredura estatica | src/uds.rs:926 | imp  
tencial efeito indireto via fluxo de controle |  
| session\_default\_relocks\_security | src/uds.rs | critica | chamadas diretas nao detectadas por varredura estatica | s  
mpacto local com potencial efeito indireto via fluxo de controle |  
| transfer\_exit\_rejects\_incomplete\_download | src/uds.rs | critica | chamadas diretas nao detectadas por varredura est

s:949 | impacto local com potencial efeito indireto via fluxo de controle |  
| bin\_path | tests/vendor\_bridge\_e2e.rs | critica | chamadas diretas nao detectadas por varredura estatica | tests/vendor\_bridge\_e2e.rs:9 | impacto local com potencial efeito indireto via fluxo de controle |  
| unique\_temp\_dir | tests/vendor\_bridge\_e2e.rs | critica | chamadas diretas nao detectadas por varredura estatica | tests/vendor\_bridge\_e2e.rs:22 | impacto local com potencial efeito indireto via fluxo de controle |  
| cat\_comm\_bridge\_probe\_and\_live\_phases\_flash\_e2e | tests/vendor\_bridge\_e2e.rs | critica | chamadas diretas nao detectadas por varredura estatica | tests/vendor\_bridge\_e2e.rs:33 | impacto local com potencial efeito indireto via fluxo de controle |  
| from\_path | src/io/allowlist.rs | alta | src/io/mock.rs ; src/leak\_physics.rs | src/io/allowlist.rs:33 ; src/io/mock.rs:1479 | pode quebrar readiness, relatorio ou integracao de hardware |  
| is\_allowed | src/io/allowlist.rs | alta | src/io/mock.rs | src/io/allowlist.rs:42 ; src/io/mock.rs:133 | pode quebrar readiness, relatorio ou integracao de hardware |  
| per\_rule\_rate\_limit | src/io/allowlist.rs | alta | chamadas diretas nao detectadas por varredura estatica | src/io/allowlist.rs:147 | impacto local com potencial efeito indireto via fluxo de controle |  
| validate | src/io/allowlist.rs | alta | chamadas diretas nao detectadas por varredura estatica | src/io/allowlist.rs:147 | impacto local com potencial efeito indireto via fluxo de controle |  
| rule\_matches\_frame | src/io/allowlist.rs | alta | chamadas diretas nao detectadas por varredura estatica | src/io/allowlist.rs:147 | impacto local com potencial efeito indireto via fluxo de controle |  
| parse\_u32\_hex\_or\_dec | src/io/allowlist.rs | alta | chamadas diretas nao detectadas por varredura estatica | src/io/allowlist.rs:147 | impacto local com potencial efeito indireto via fluxo de controle |  
| parse\_mask | src/io/allowlist.rs | alta | tests/io\_mock.rs | src/io/allowlist.rs:146 ; tests/io\_mock.rs:190 | pode quebrar readiness, relatorio ou integracao de hardware |  
| match\_hex\_pattern | src/io/allowlist.rs | alta | chamadas diretas nao detectadas por varredura estatica | src/io/allowlist.rs:147 | impacto local com potencial efeito indireto via fluxo de controle |  
| can\_rule\_matches\_mask | src/io/allowlist.rs | alta | chamadas diretas nao detectadas por varredura estatica | src/io/allowlist.rs:147 | impacto local com potencial efeito indireto via fluxo de controle |  
| serial\_rule\_matches\_pattern | src/io/allowlist.rs | alta | chamadas diretas nao detectadas por varredura estatica | src/io/allowlist.rs:209 | impacto local com potencial efeito indireto via fluxo de controle |  
| write\_intent\_enabled | src/io/hw.rs | alta | src/io/mock.rs ; tests/io\_mock.rs | src/io/hw.rs:223 ; src/io/mock.rs:140 | pode quebrar readiness, relatorio ou integracao de hardware |  
| connect\_ecm | src/io/live\_runner.rs | alta | src/io/production\_program.rs ; src/main.rs | src/io/live\_runner.rs:174 ; src/main.rs:171 ; src/main.rs:2655 | pode quebrar readiness, relatorio ou integracao de hardware |  
| live\_preflight\_ecm | src/io/live\_runner.rs | alta | src/io/production\_program.rs | src/io/live\_runner.rs:793 ; src/main.rs:271 | pode quebrar readiness, relatorio ou integracao de hardware |  
| check\_live\_channel\_policy | src/io/live\_runner.rs | alta | chamadas diretas nao detectadas por varredura estatica | src/io/live\_runner.rs:1216 | impacto local com potencial efeito indireto via fluxo de controle |  
| on\_rate\_limited | src/io/metrics.rs | alta | src/io/mock.rs | src/io/metrics.rs:37 ; src/io/mock.rs:149 | pode quebrar readiness, relatorio ou integracao de hardware |  
| inject\_disconnect | src/io/mock.rs | alta | chamadas diretas nao detectadas por varredura estatica | src/io/mock.rs:302 | impacto local com potencial efeito indireto via fluxo de controle |  
| clear\_disconnect | src/io/mock.rs | alta | chamadas diretas nao detectadas por varredura estatica | src/io/mock.rs:302 | impacto local com potencial efeito indireto via fluxo de controle |  
| read\_frame | src/io/mock.rs | alta | src/io/hw.rs ; src/io/live\_runner.rs ; src/io/live\_runner.rs ; src/io/live\_runner.rs ; src/io/live\_runner.rs | src/io/mock.rs:61 ; src/io/hw.rs:288 ; src/io/live\_runner.rs:148 ; src/io/live\_runner.rs:302 ; src/io/live\_runner.rs:702 ; src/io/live\_runner.rs:897 | pode quebrar readiness, relatorio ou integracao de hardware |  
| try\_read\_frame | src/io/mock.rs | alta | src/io/hw.rs ; src/io/serial\_adapter.rs ; src/io/socketcan\_adapter.rs ; src/main.rs | src/io/mock.rs:89 ; src/io/hw.rs:289 ; src/io/serial\_adapter.rs:61 ; src/io/socketcan\_adapter.rs:80 ; src/main.rs:261 | pode quebrar readiness, relatorio ou integracao de hardware |  
| send\_frame | src/io/mock.rs | alta | src/io/hw.rs ; src/io/live\_runner.rs ; src/io/live\_runner.rs ; src/io/serial\_adapter.rs ; src/io/socketcan\_adapter.rs | src/io/mock.rs:114 ; src/io/hw.rs:290 ; src/io/live\_runner.rs:200 ; src/io/serial\_adapter.rs:69 ; src/io/socketcan\_adapter.rs:89 ; src/io/socketcan\_adapter.rs:124 | pode quebrar readiness, relatorio ou integracao de hardware |  
| default | src/io/production\_program.rs | alta | src/autonomous.rs ; src/bin/cat\_comm\_bridge.rs ; src/bin/cat\_comm\_bridge.rs ; src/bin/cat\_comm\_bridge.rs ; src/camera.rs ; src/camera.rs | src/io/production\_program.rs:69 ; src/autonomous.rs:184 ; src/bin/cat\_comm\_bridge.rs:184 ; src/boot\_sequence.rs:178 ; src/camera.rs:142 ; src/camera.rs:246 | pode quebrar readiness, relatorio ou integracao de hardware |  
| run\_sim\_baseline\_check | src/io/production\_program.rs | alta | chamadas diretas nao detectadas por varredura estatica | src/io/production\_program.rs:568 | impacto local com potencial efeito indireto via fluxo de controle |  
| run\_sim\_connectivity\_check | src/io/production\_program.rs | alta | chamadas diretas nao detectadas por varredura estatica | src/io/production\_program.rs:587 | impacto local com potencial efeito indireto via fluxo de controle |  
| run\_sim\_preflight\_check | src/io/production\_program.rs | alta | chamadas diretas nao detectadas por varredura estatica | src/io/production\_program.rs:731 | impacto local com potencial efeito indireto via fluxo de controle |  
| run\_sim\_stress\_check | src/io/production\_program.rs | alta | chamadas diretas nao detectadas por varredura estatica | src/io/production\_program.rs:763 | impacto local com potencial efeito indireto via fluxo de controle |  
| hex\_sha | src/io/production\_program.rs | alta | chamadas diretas nao detectadas por varredura estatica | src/io/production\_program.rs:784 | impacto local com potencial efeito indireto via fluxo de controle |  
| now\_ms | src/io/production\_program.rs | alta | src/bin/cat\_comm\_bridge.rs ; src/bin/cat\_comm\_bridge.rs ; src/io/hw.rs ; src/io/live\_runner.rs ; src/io/live\_runner.rs | src/io/production\_program.rs:792 ; src/bin/cat\_comm\_bridge.rs:97 ; src/bin/cat\_comm\_bridge.rs:362 ; src/io/hw.rs:353 ; src/io/hw.rs:400 ; src/io/live\_runner.rs:200 ; src/io/live\_runner.rs:373 | pode quebrar readiness, relatorio ou integracao de hardware |



| new | src/io/rate\_limiter.rs | alta | src/adas.rs ; src/autonomous.rs ; src/autonomous.rs ; src/autonomous.rs ; src/autonomous.rs | src/io/rate\_limiter.rs:13 ; src/adas.rs:67 ; src/autonomous.rs:185 ; src/autonomous.rs:190 ; src/autonomous.rs:545 ; src/autonomous.rs:570 | pode quebrar readiness, relatorio ou integracao de hardware |

| try\_take | src/io/rate\_limiter.rs | alta | chamadas diretas nao detectadas por varredura estatica | src/io/rate\_limiter.rs:13 | impacto local com potencial efeito indireto via fluxo de controle |

| refill | src/io/rate\_limiter.rs | alta | chamadas diretas nao detectadas por varredura estatica | src/io/rate\_limiter.rs:13 | impacto local com potencial efeito indireto via fluxo de controle |

| new | src/io/rate\_limiter.rs | alta | src/adas.rs ; src/autonomous.rs ; src/autonomous.rs ; src/autonomous.rs ; src/autonomous.rs | src/io/rate\_limiter.rs:49 ; src/adas.rs:67 ; src/autonomous.rs:185 ; src/autonomous.rs:190 ; src/autonomous.rs:545 ; src/autonomous.rs:570 | pode quebrar readiness, relatorio ou integracao de hardware |

| check\_can | src/io/rate\_limiter.rs | alta | src/io/mock.rs | src/io/rate\_limiter.rs:57 ; src/io/mock.rs:141 | pode quebrar readiness, relatorio ou integracao de hardware |

| check\_serial | src/io/rate\_limiter.rs | alta | src/io/mock.rs | src/io/rate\_limiter.rs:70 ; src/io/mock.rs:142 | pode quebrar readiness, relatorio ou integracao de hardware |

| limiter\_blocks\_when\_empty | src/io/rate\_limiter.rs | alta | chamadas diretas nao detectadas por varredura estatica | src/io/rate\_limiter.rs:80 | impacto local com potencial efeito indireto via fluxo de controle |

| aggregate\_group\_report | src/leak\_physics.rs | alta | chamadas diretas nao detectadas por varredura estatica | src/leak\_physics.rs:1654 ; src/lib.rs:112 | impacto local com potencial efeito indireto via fluxo de controle |

| export\_calibration\_report\_json | src/leak\_physics.rs | alta | src/lib.rs | src/leak\_physics.rs:1670 ; src/lib.rs:1131 | pode quebrar readiness, relatorio ou integracao de hardware |

| export\_calibration\_report\_csv | src/leak\_physics.rs | alta | src/lib.rs | src/leak\_physics.rs:1670 ; src/lib.rs:1131 | pode quebrar readiness, relatorio ou integracao de hardware |

| report\_captures\_pressure\_band\_and\_safe\_window | src/leak\_physics.rs | alta | chamadas diretas nao detectadas por varredura estatica | src/leak\_physics.rs:2030 | impacto local com potencial efeito indireto via fluxo de controle |

| calibration\_mode\_ingests\_csv\_and\_exports\_reports | src/leak\_physics.rs | alta | chamadas diretas nao detectadas por varredura estatica | src/leak\_physics.rs:2107 | impacto local com potencial efeito indireto via fluxo de controle |

| export\_leak\_report\_json | src/lib.rs | alta | src/main.rs ; tests/leak\_system\_integration.rs | src/lib.rs:1053 ; src/main.rs:1053 | pode quebrar readiness, relatorio ou integracao de hardware |

| export\_leak\_report\_csv | src/lib.rs | alta | src/main.rs ; tests/leak\_system\_integration.rs | src/lib.rs:1060 ; src/main.rs:1060 | pode quebrar readiness, relatorio ou integracao de hardware |

| export\_leak\_calibration\_report\_json | src/lib.rs | alta | src/main.rs | src/lib.rs:1118 ; src/main.rs:1142 | pode quebrar readiness, relatorio ou integracao de hardware |

| export\_leak\_calibration\_report\_csv | src/lib.rs | alta | src/main.rs | src/lib.rs:1126 ; src/main.rs:1130 | pode quebrar readiness, relatorio ou integracao de hardware |

| pick\_save\_path | src/main.rs | alta | chamadas diretas nao detectadas por varredura estatica | src/main.rs:542 | impacto local com potencial efeito indireto via fluxo de controle |

| leak\_ascii\_cad | src/main.rs | alta | chamadas diretas nao detectadas por varredura estatica | src/main.rs:6360 | impacto local com potencial efeito indireto via fluxo de controle |

| fmt | src/v2x\_telematics.rs | alta | src/adas.rs ; src/autonomous.rs ; src/autonomous.rs ; src/autonomous.rs ; src/autonomous.rs | src/v2x\_telematics.rs:322 ; src/adas.rs:56 ; src/autonomous.rs:16 ; src/autonomous.rs:39 ; src/autonomous.rs:83 ; src/autonomous.rs:173 | pode quebrar readiness, relatorio ou integracao de hardware |

| write\_test\_allowlist | tests/io\_mock.rs | alta | chamadas diretas nao detectadas por varredura estatica | tests/io\_mock.rs:97 | impacto local com potencial efeito indireto via fluxo de controle |

| mock\_adapter\_allows\_write\_when\_policy\_satisfied | tests/io\_mock.rs | alta | chamadas diretas nao detectadas por varredura estatica | tests/io\_mock.rs:97 | impacto local com potencial efeito indireto via fluxo de controle |

| mock\_adapter\_dry\_run\_logs\_without\_physical\_tx | tests/io\_mock.rs | alta | chamadas diretas nao detectadas por varredura estatica | tests/io\_mock.rs:125 | impacto local com potencial efeito indireto via fluxo de controle |

| mock\_adapter\_rate\_limit\_blocks\_burst | tests/io\_mock.rs | alta | chamadas diretas nao detectadas por varredura estatica | tests/io\_mock.rs:157 | impacto local com potencial efeito indireto via fluxo de controle |

| allowlist\_parse\_mask\_accepts\_hex | tests/io\_mock.rs | alta | chamadas diretas nao detectadas por varredura estatica | tests/io\_mock.rs:189 | impacto local com potencial efeito indireto via fluxo de controle |

| e2e\_hydraulic\_plus\_ac\_stress\_generates\_reports | tests/leak\_system\_integration.rs | alta | chamadas diretas nao detectadas por varredura estatica | tests/leak\_system\_integration.rs:4 | impacto local com potencial efeito indireto via fluxo de controle |

</details>

### ## 3. Roadmap de Refatoracao Segura por Ondas

#### ### Wave 1 - Blindagem de Contratos de Protocolo

- Escopo: src/io/live\_runner.rs, src/io/vendor\_cat\_comm.rs, src/uds.rs
- Mudancas: centralizar validacao de SID positivo, tipar NRC, consolidar parse de DIDs
- Checklist de testes:
  - [ ] cargo check
  - [ ] cargo test --workspace
  - [ ] cargo test --features vendor-windows --test vendor\_bridge\_e2e

#### ### Wave 2 - Refino de Abstracao de Hardware

- Escopo: src/io/hw.rs, src/io/socketcan\_adapter.rs, src/io/serial\_adapter.rs
- Mudancas: contratos de capability com traits menores, erros padronizados por camada
- Checklist de testes:

- [ ] cargo check --features vendor-windows
- [ ] cargo test --workspace
- [ ] cargo test --test io\_mock

### ### Wave 3 - Estabilizacao de Schema de Relatorio

- Escopo: src/io/production\_program.rs, src/bin/simulator\_cli.rs, docs/ECM-Data.md
- Mudancas: versionamento de schema, compat adapters para consumidores antigos
- Checklist de testes:
  - [ ] cargo check
  - [ ] cargo test --workspace
  - [ ] powershell -NoProfile -ExecutionPolicy Bypass -File scripts/run\_windows\_template\_e2e.ps1

### ### Wave 4 - Performance e Observabilidade

- Escopo: src/main.rs, src/observability.rs, src/io/metrics.rs
- Mudancas: reduzir jitter de render e padronizar eventos de telemetria
- Checklist de testes:
  - [ ] cargo check
  - [ ] cargo test --workspace

### ### Wave 5 - Harden de CI e Evidencia

- Escopo: .github/workflows/\*.yaml, scripts/\*.ps1, scripts/\*.sh
- Mudancas: gates obrigatorios por risco, artefatos de report/hash sempre anexados
- Checklist de testes:
  - [ ] cargo check
  - [ ] cargo test --workspace
  - [ ] cargo test --features vendor-windows --test vendor\_bridge\_e2e

### ## Cobertura e Limites

- Esta edicao prioriza funcoes de alto impacto e rastreabilidade de chamadas por varredura estatica.
- Chamadas dinamicas e dispatch indireto podem nao aparecer na tabela X->Y.